

Houlin Wang

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PROFILE

Biomedical Engineering undergraduate specializing in instrumentation, with a Computer Science minor at Zhejiang University. Trained at the intersection of neural data analysis, medical AI, biosensor instrumentation, and embedded biomedical systems, with experience in iEEG analysis, medical image modeling, circuit-level sensing hardware, and embedded system development.

RESEARCH INTERESTS

Neural engineering; biomedical signal processing; AI for healthcare; embedded biomedical systems; reproducible analysis pipelines for complex biomedical data.

EDUCATION

Zhejiang University

B.Eng. in Biomedical Engineering (Instrumentation); Minor in Computer Science

Sep. 2023 – Present

GPA: 4.58/5.0 (91.25/100)

- Interdisciplinary training across biomedical engineering, computer science, embedded systems, signal processing, and medical foundations.
- Completed 188 credits in the first three undergraduate years, exceeding the 160-credit graduation requirement.
- Selected major and technical coursework: Introduction to Pattern Analysis and Machine Learning, Biomedical Signal Processing, Biomedical Image Processing, Biomedical Imaging Technology, Biomedical Sensors and Measurement, Instrument System Design, Embedded Systems, Microcomputer Theory and Application, Electric Circuit and Electronic Technology I–II, Fundamentals of Data Structures, Computer Organization, Digital Logic Design, Hardware Description Language, Probability and Mathematical Statistics, Engineering Physiology.
- Medical preparation: audited Regional Anatomy and participated in anatomy laboratory sessions; enrolled in an English-taught Systematic Anatomy course.

RESEARCH EXPERIENCE

Medial Temporal Lobe iEEG Analysis of Working Memory and Hippocampal Ripples

Aug. 2025 – Present

Undergraduate Researcher, Qiushi Academy for Advanced Studies, Zhejiang University

- Developed a reproducible MATLAB pipeline for ripple detection and trial-level working-memory analysis across multi-subject, multi-session medial temporal lobe iEEG data.
- Characterized ripple dynamics across task phases, memory loads, and behavioral outcomes, and analyzed co-ripple patterns and event-triggered activity using permutation tests, mixed-effects models, and BH-FDR correction.
- Established rigorous cross-session trial mapping and event-alignment rules across subjects, sessions, trials, brain regions, and hemispheres to ensure data integrity and valid cross-region event matching.
- Extended the analysis from ripple co-occurrence to trial-specific encoding-pattern reinstatement, constructing hippocampal joint-reinstatement metrics and evaluating correctness-related differences during memory maintenance with matched non-ripple controls and robustness checks.

Deep Learning-Based Corneal Confocal Microscopy Diagnosis System

Apr. 2025 – May 2026

Undergraduate Researcher, Zhejiang University

- Curated and standardized corneal confocal microscopy images and pixel-level masks, including lesion screening, annotation quality control, and image-mask formatting.
- Conducted classification and segmentation experiments involving image preprocessing, CLIP-based encoding, multi-scale feature fusion, model evaluation, and error analysis.
- Implemented backend modules for data management, model inference, and front-end/back-end communication, and supported cross-platform model deployment.
- The resulting models achieved 96.85% average accuracy and 98.2% average AUC across 11 normal and pathological image categories, while the segmentation model achieved a mean Dice score of 82.27%.

ENGINEERING AND COMPETITION EXPERIENCE

National Undergraduate Electronics Design Contest Regional Contest and TI Cup Analog Electronic System Design Challenge

Jul. 2026

Competition Team Member, Zhejiang University

- Worked on analog electronic system design, including signal conditioning, amplification, power output, and ADC-related modules.
- Participated in module integration, inter-module interfacing, and system-level debugging across analog and embedded subsystems.
- Developed and debugged STM32/MSPM0 code for control and data acquisition; also contributed to an MSPM0/K230 intelligent vehicle project involving embedded control and vision-based AI recognition.

ZJU TruSense Biosensor Design Competition Team

Oct. 2024 – Sep. 2025

Competition Team Member, Zhejiang University / SensUs Student Competition

- Developed the instrumentation subsystem of a portable creatinine-sensing platform for acute kidney injury monitoring.
- Designed and implemented a three-electrode electrochemical measurement circuit for acquiring creatinine-dependent sensor signals.
- Led schematic design and five PCB iterations for an MSP430FR2433-based system, reducing the hardware toward a coin-sized form factor and improving layout reliability.
- Performed soldering, hardware debugging, and prototype-level troubleshooting during system integration.
- Participated in embedded code development for the instrumentation subsystem, connecting circuit implementation with sensing and acquisition requirements.

PUBLICATIONS

- Chentao Jiang, Ruiqi Yang, **Houlin Wang**, et al. *A multi-disease in vivo confocal microscopy dataset for AI-based corneal disease diagnosis and research*. Submitted in 2025; under review.

HONORS AND AWARDS

Scholarships

- Zhejiang Provincial Government Scholarship, 2024–2025.
- Second-Class Scholarship, Zhejiang University, 2024–2025.
- Third-Class Scholarship, Zhejiang University, 2023–2024.

Competition Awards

- Gold Medal, SynBio Challenges 2025, Red Track: Health, Medical & Treatment, Team TruSense, Aug. 2025.
- National Bronze Award (Final Round) and Provincial First Prize (Provincial Round), 11th National College Student Medical Innovation Competition and “Belt and Road” International Competition, Interdisciplinary Track, Jun.–Aug. 2025.
- Final Medal; Public Inspiration 1st Prize; Innovation 3rd Prize, SensUs Student Competition 2025: Kidney Failure, Team TruSense, Aug. 2025.
- Third Prize, 10th National College Student Biomedical Engineering Innovation Design Competition, Jul. 2025.

SKILLS

Programming	Python, MATLAB, C/C++, Java, Verilog
Data & Machine Learning	PyTorch, NumPy, SciPy, CLIP-Based Vision-Language Models
Neural Signal Analysis	FieldTrip, FMA Toolbox, EEGLAB
Embedded Systems	STM32, MSPM0, MSP430, ESP32, 8051
Hardware & Toolchains	EasyEDA, Vivado, Keil MDK-ARM, Keil C51, CCS, STM32CubeMX, PlatformIO
Documentation	Git, Markdown, LaTeX, Typst

LANGUAGES

Chinese	Native
English	IELTS 8.0 (R9.0 / L9.0 / W7.0 / S6.5); TOEFL iBT 101 (R29 / L24 / S25 / W23)
French	Basic conversational proficiency; TFU-4 96/100

SERVICE AND ADDITIONAL INFORMATION

- Volunteer Teacher, Western China Education Volunteer Program, Hongya, Sichuan, Jul.–Aug. 2024; served as a home-room teacher and instructor.